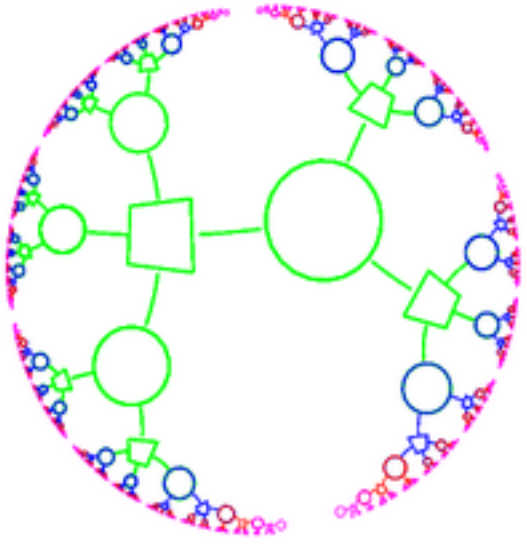


# Information theory, pattern recognition, and neural networks


- 1 Noisy-channel coding

Source coding (Data compression)

- 2 Information content, entropy
- 3 Typicality and the source coding theorem
- 4 Symbol codes


www.inference.phy.cam.ac.uk/itprnn/

# How to compress a redundant file?

```
0000000000000000010010010000000000000000010000100010100000
0001010000000000000000000010000000010000000000000100000110
0000100000000000000010000000000100001000001000000000000000
0000100000010000000000100000000000000000000000010000000000
0000000100010000001000000100100000000001000000010000000000
0001001000010101000000110000100000000000100000000000000000
0000000000000000000000010100000010000000000001000101000001
1000000000100000000000000000000000000000000000000000000000
0100000000000010000000000000001000010000000001101100000000
1010001010000000000000000000000000001000101000010000010000
```

e.g., $N = 1000$ tosses of a bent coin with $p_1 = 0.1$

# How to measure information content?

**Claims:**

1. The *Shannon information content* of an outcome

$$h(x=a_i) = \log_2 \frac{1}{P(x=a_i)}$$

is a sensible measure of information content.

2. The entropy

$$H(X) = \sum_x P(x) \log_2 \frac{1}{P(x)}$$

is a sensible measure of expected information content.

3. Source coding theorem –
$N$ outcomes from a source $X$ can be compressed into roughly $NH(X)$ bits.

3. Source coding theorem –

$N$ outcomes from a source $X$ can be compressed into roughly $NH(X)$ bits.

Proved by counting the typical set

When a source $X$
produces $N$ independent outcomes

$$\mathbf{x} = x_1 x_2 \ldots x_N$$

this string is very likely to be one of the

$$\sim 2^{NH(X)} \textit{ typical} \text{ outcomes}$$

all of which have probability $\sim 2^{-NH(X)}$

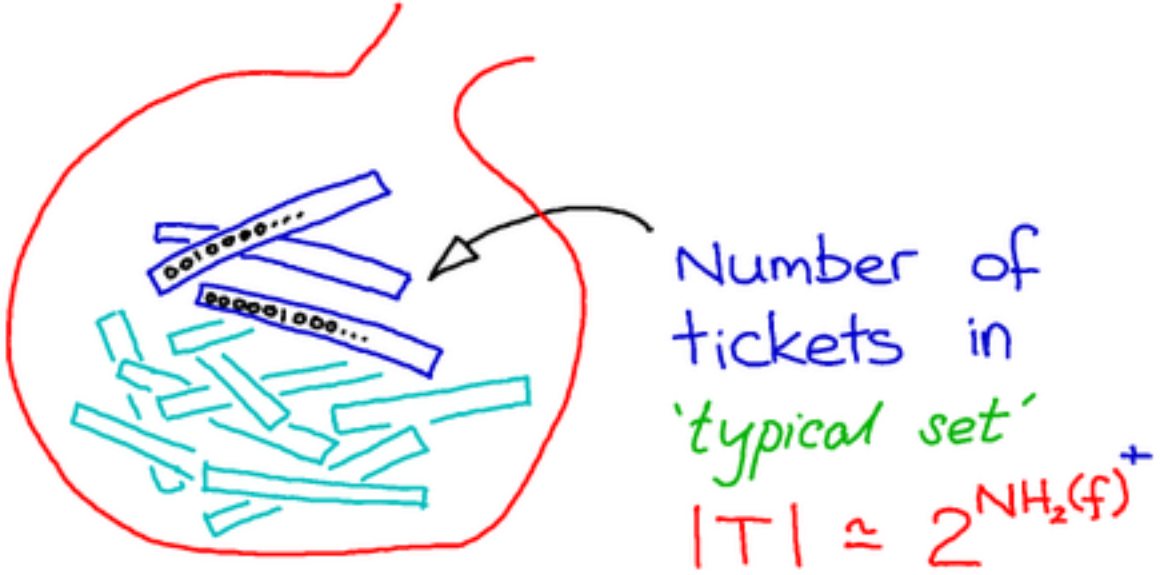
# Example: the bent coin

```
000000000000000001001001000000000000000000100001000101000000
000101000000000000000000001000000001000000000000100000110
000010000000000000010000000001000010000010000000000000000
000010000001000000000010000000000000000000000010000000000
000000010001000000100000010010000000000010000000010000000000
000100100001010100000011000010000000000010000000000000000000
000000000000000000000001010000001000000000000100010100001
100000000010000000000000000000000000000000000000000000000000
010000000000001000000000000000010000100000000011011000000
101000101000000000000000000000000000010001010000100001000000
```

# How we won the bent coin lottery

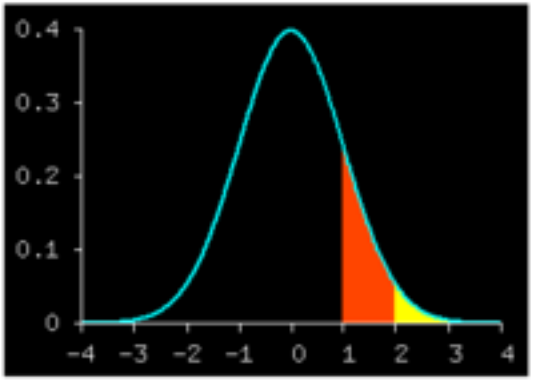
Probability of '1' = f

To have a 99.99% chance of winning,
we bought all the typical tickets



# Example: the bent coin

```
00000000000000000100100100000000000000000100001000101000000
00010100000000000000000000010000000010000000000000100000110
00001000000000000001000000000010000100000100000000000000000
00001000000100000000001000000000000000000000000010000000000
00000001000100000010000001001000000000001000000001000000000
00010010000101010000011000010000000000010000000000000000000
00000000000000000000000010100000100000000000010001010000 1
10000000001000000000000000000000000000000000000000000000000
01000000000000100000000000000010000100000000011011000000 0
10100010100000000000000000000000000010001010000100001000000
```

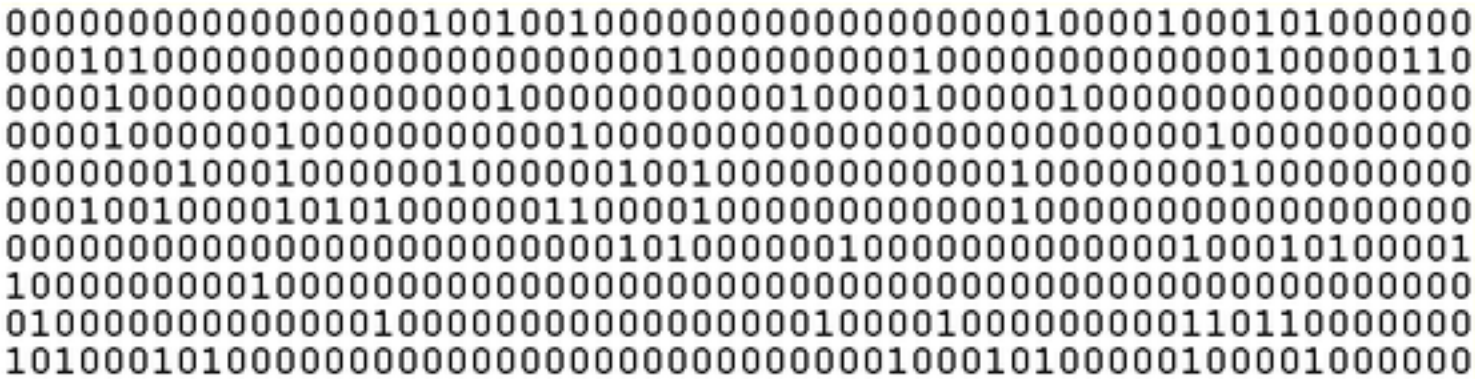



Gaussian distribution

| | |
|---|---|
| $P(x > 1\sigma)$ | 16% |
| $P(x > 2\sigma)$ | 2.3% |
| $P(x > 2.3\sigma)$ | 1% |
| $P(x > 3\sigma)$ | 0.13% |

# How to compress a redundant file,

## practically?



Emma Woodhouse, handsome, clever, and rich, with a comfortable home and happy disposition, seemed to unite some of the best blessings of existence; and had lived nearly twenty one years in the world with very little to distress or vex her. She was the youngest of the two daughters of a most affectionate, indulgent father; and had, in consequence of her sister's marriage, been mistress of his house from a very early period. Her mother had died too long ago for her to have more than an indistinct remembrance of her caresses; and her place had been supplied by an excellent woman as governess, who had fallen little short of a mother in affection. Sixteen years had Miss Taylor been in Mr Woodhouse's family, less as a governess than a friend, very

| $i$ | $a_i$ | $p_i$ |
|---|---|---|
| 1 | a | 0.0575 |
| 2 | b | 0.0128 |
| 3 | c | 0.0263 |
| 4 | d | 0.0285 |
| 5 | e | 0.0913 |
| 6 | f | 0.0173 |
| 7 | g | 0.0133 |
| 8 | h | 0.0313 |
| 9 | i | 0.0599 |
| 10 | j | 0.0006 |
| 11 | k | 0.0084 |
| 12 | l | 0.0335 |
| 13 | m | 0.0235 |
| 14 | n | 0.0596 |
| 15 | o | 0.0689 |
| 16 | p | 0.0192 |
| 17 | q | 0.0008 |
| 18 | r | 0.0508 |
| 19 | s | 0.0567 |
| 20 | t | 0.0706 |
| 21 | u | 0.0334 |
| 22 | v | 0.0069 |
| 23 | w | 0.0119 |
| 24 | x | 0.0073 |
| 25 | y | 0.0164 |
| 26 | z | 0.0007 |
| 27 | – | 0.1928 |




```
a 0000
b 001000
c 00101
d 10000
e 1100
f 111000
g 001001
h 10001
i 1001
j 1101000000
k 1010000
l 11101
m 110101
n 0001
o 1011
p 111001
q 110100001
r 11011
s 0011
t 1111
u 10101
v 11010001
w 1101001
x 1010001
y 101001
z 1101000001
– 01
```

$C(x) \in \{0,1\}$

a binary string

$$\{0,1\}^2 = \{00, 01, 10, 11\}$$

symbol $x \longrightarrow C(x) \in \{0,$

a binary string

$\{0, 1$

we encode $X_1 X_2 X_3 \ldots X_N$

as $C(X_1)C(X_2)C(X_3)\ldots C(X_N)$

a 1

b 010

c 1011

cabb → 1011010010

## THEORY

How well might a symbol code perform?

## PRACTICAL

How to make optimal symbol codes?

Expected length of a code $C$ for an ensemble $X$

$$L(C,X) = \sum_i p_i l_i$$

$$A_x = \{a, b, c, d\}$$

$$P_x = \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right\}$$

$$H(X) = \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \cdot 3 + \frac{1}{8} \cdot 3$$

$$= \underline{1\,3/4 \text{ bits}}$$

# Rules

- Decode correctly!
  UNIQUELY DECODEABLE
  for any $\underline{x}, \underline{y}$ $[\underline{x} \neq \underline{y}]$: $C(\underline{x}) \neq C(\underline{y})$
- Easy to decode
- Small $L(C, X)$

LE

$\neq y]$: $C(x) \neq C(y)$

a

b

c

$C(a_i)$

$C(x) = C(x_1)\,C(x_2)\ldots C(x_N)$

$C_1 = \{1000, 0100, 0010, 0001\}$

$L = 4$

$C_2 = \{1, 01, 001, 0001\}$

$L = 1.875 = 1\frac{7}{8}$

$\{0, 1, 0, 1\}$

$L = 1$

Not UD

all codewords must be different !

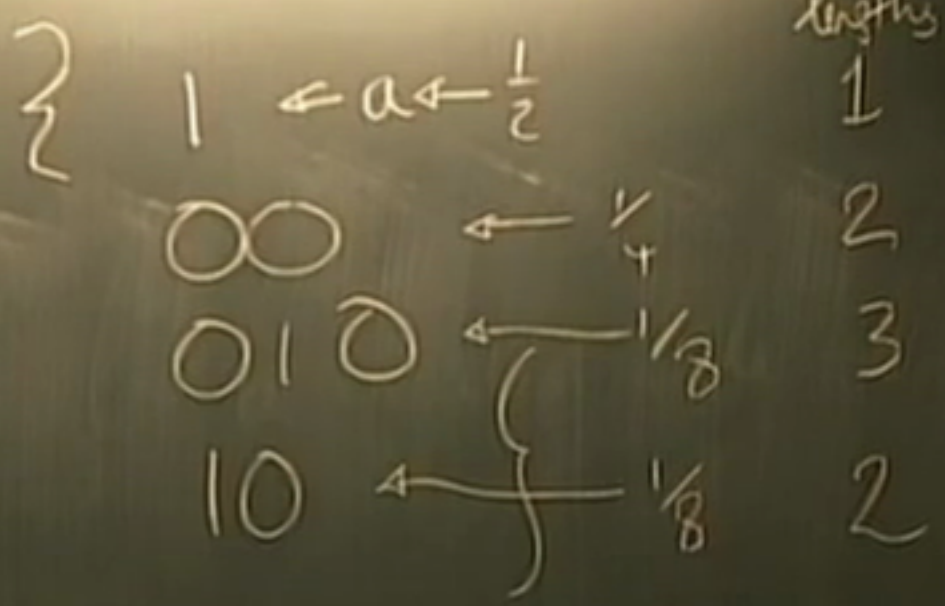
$C_1 = \{$

$\forall\ [\underline{x} \neq \underline{y}]: \quad C(\underline{x}) \neq C(\underline{y})$

$C(a_i)$

$C(\underline{x}) = C(x_1)\,C(x_2)\,..\,C(x_N)$

$L =$




$L = 15/8$

$L = \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{8} \times 2$

$C_4$: caab

c(caab)

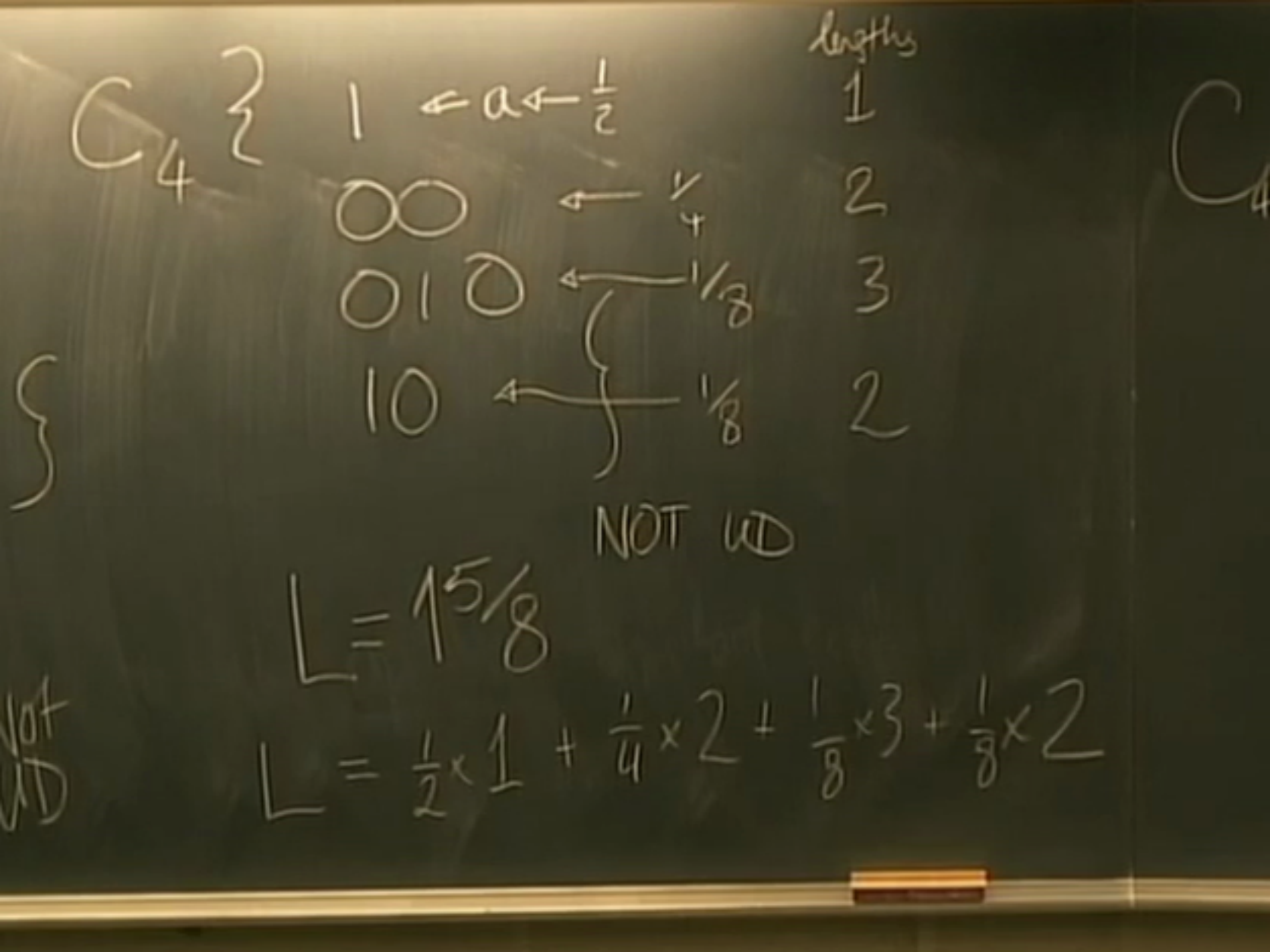
= 0101100...

c caab

a d

$\underline{x}$ = dc 10010

$\underline{y}$ = abd 10010


lengths
$C_4$ { 1 ← a ← 1/2 1
00 ← 1/4 2
010 ← 1/8 3
10 ← 1/8 2
}
NOT UD
L = 15/8
Not UD
$L = \frac{1}{2} \times 1 + \frac{1}{4} \times 2 + \frac{1}{8} \times 3 + \frac{1}{8} \times 2$

$C_5 = \{ 00, 01, 10, 11 \}$

$L = 2$

100...

a b

a d

= dc 10010

= abd 10010

$C_2$

$C_6 = \{$ 1 — 1

01 — 2

001 — 3

000 — 3 $\}$

$L = 1\ 3\ 4 = H$

caab → 0011101

THIS IS A PREFIX CODE

NO codeword is a prefix of any other

1 1

01 2

001 3

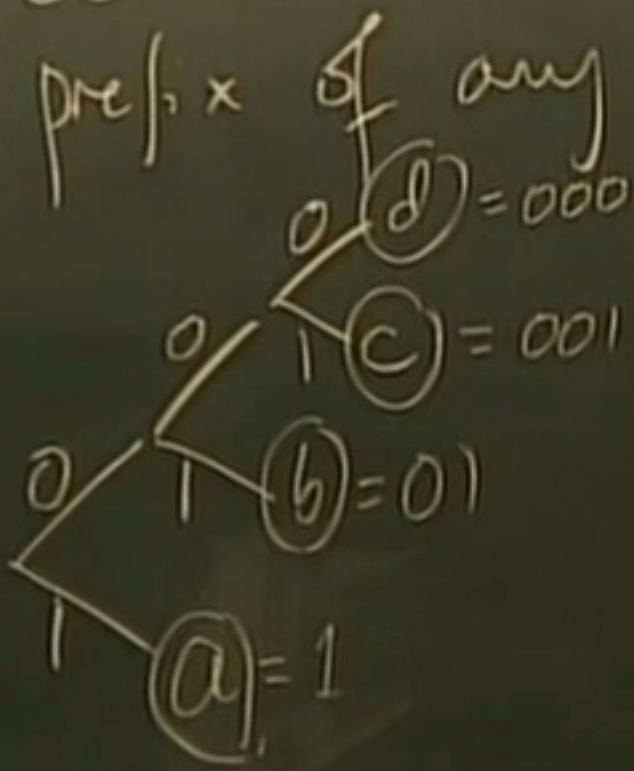
000 } 3

= 13/4 = H

→ 00111101

# Code

No codeword is a prefix of any other.



$C_7 = \{ 1, 10, 100, 000 \}$

Not a prefix code

$C_7 = \{$ 1, 10, 100, 000 $\}$

$A_x = \{a, b$

$P_x = \{ \frac{1}{2}, $

$h_i \xrightarrow{x} 1$

$H(X) = \frac{1}{2} \times 1$

$= 1^3$

Not a prefix code

10000000006000

dea ...

Rank by $p$ then

| | Codewords |
|---|---|
| $p_1$ | 1 |
| $p_2$ | 01 |
| | 001 |
| | 0001 |
| | ⋮ |
| $p_{I-1}$ | 00001 |
| $p_I$ | 00000 |

$I-1$

$p_1$ 1 1 a, b, c, d

$p_2$ 01 2 $\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$ $\frac{1}{4}$

001 3

0001

$p_{I-1}$ 00001

$p_I$ 00000 3

$\longleftrightarrow$

$I-1$

$A_x = \{a$

$P_x = \{ \frac{1}{2}$

$C_6$

$h_i \xrightarrow{x} 1$

$L = \frac{9}{4}$

$H(X) = \frac{1}{2}$

$= 2\frac{1}{4}$ bits

If all codewords have length

| $\ell$ | 1 | 2 | 3 | $\ell$ |
|---|---|---|---|---|
| # | 2 | 4 | 8 | $2^{\ell}$ |

# The symbol-code supermarket

<table>
<tr><td rowspan="8">0</td><td rowspan="4">00</td><td rowspan="2">000</td><td>0000</td></tr>
<tr><td>0001</td></tr>
<tr><td rowspan="2">001</td><td>0010</td></tr>
<tr><td>0011</td></tr>
<tr><td rowspan="4">01</td><td rowspan="2">010</td><td>0100</td></tr>
<tr><td>0101</td></tr>
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<tr><td>0111</td></tr>
<tr><td rowspan="8">1</td><td rowspan="4">10</td><td rowspan="2">100</td><td>1000</td></tr>
<tr><td>1001</td></tr>
<tr><td rowspan="2">101</td><td>1010</td></tr>
<tr><td>1011</td></tr>
<tr><td rowspan="4">11</td><td rowspan="2">110</td><td>1100</td></tr>
<tr><td>1101</td></tr>
<tr><td rowspan="2">111</td><td>1110</td></tr>
<tr><td>1111</td></tr>
</table>

# The symbol-code supermarket

| | | | |
|---|---|---|---|
| 0 | 00 | 000 | 0000 |
| | | | 0001 |
| | | 001 | 0010 |
| | | | 0011 |
| | 01 | 010 | 0100 |
| | | | 0101 |
| | | 011 | 0110 |
| | | | 0111 |
| 1 | 10 | 100 | 1000 |
| | | | 1001 |
| | | 101 | 1010 |
| | | | 1011 |
| | 11 | 110 | 1100 |
| | | | 1101 |
| | | 111 | 1110 |
| | | | 1111 |

If a code is UD then

$$\sum_i 2^{-l_i} \leq 1$$

Kraft inequality

# The symbol-code supermarket

| | | | |
|---|---|---|---|
| 0 | 00 | 000 | 0000 |
| | | | 0001 |
| | | 001 | 0010 |
| | | | 0011 |
| | 01 | 010 | 0100 |
| | | | 0101 |
| | | 011 | 0110 |
| | | | 0111 |
| 1 | 10 | 100 | 1000 |
| | | | 1001 |
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| | | | 1011 |
| | 11 | 110 | 1100 |
| | | | 1101 |
| | | 111 | 1110 |
| | | | 1111 |

# The symbol-code supermarket

| | | | |
|---|---|---|---|
| 0 | 00 | 000 | 0000 |
| | | | 0001 |
| | | 001 | 0010 |
| | | | 0011 |
| | 01 | 010 | 0100 |
| | | | 0101 |
| | | 011 | 0110 |
| | | | 0111 |
| 1 | 10 | 100 | 1000 |
| | | | 1001 |
| | | 101 | 1010 |
| | | | 1011 |
| | 11 | 110 | 1100 |
| | | | 1101 |
| | | 111 | 1110 |
| | | | 1111 |

# The symbol-code supermarket

| | | | |
|---|---|---|---|
| 0 | 00 | 000 | 0000 |
| | | | 0001 |
| | | 001 | 0010 |
| | | | 0011 |
| | 01 | 010 | 0100 |
| | | | 0101 |
| | | 011 | 0110 |
| | | | 0111 |
| 1 | 10 | 100 | 1000 |
| | | | 1001 |
| | | 101 | 1010 |
| | | | 1011 |
| | 11 | 110 | 1100 |
| | | | 1101 |
| | | 111 | 1110 |
| | | | 1111 |

# The symbol-code supermarket

| | | | |
|---|---|---|---|
| 0 | 00 | 000 | 0000 |
| | | | 0001 |
| | | 001 | 0010 |
| | | | 0011 |
| | 01 | 010 | 0100 |
| | | | 0101 |
| | | 011 | 0110 |
| | | | 0111 |
| 1 | 10 | 100 | 1000 |
| | | | 1001 |
| | | 101 | 1010 |
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| | 11 | 110 | 1100 |
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<tr><td>0011</td></tr>
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<tr><td rowspan="2">011</td><td>0110</td></tr>
<tr><td>0111</td></tr>
<tr><td rowspan="8">1</td><td rowspan="4">10</td><td rowspan="2">100</td><td>1000</td></tr>
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<tr><td>1111</td></tr>
</table>

# The symbol-code supermarket

| | | | |
|---|---|---|---|
| 0 | 00 | 000 | 0000 |
| | | | 0001 |
| | | 001 | 0010 |
| | | | 0011 |
| | 01 | 010 | 0100 |
| | | | 0101 |
| | | 011 | 0110 |
| | | | 0111 |
| 1 | 10 | 100 | 1000 |
| | | | 1001 |
| | | 101 | 1010 |
| | | | 1011 |
| | 11 | 110 | 1100 |
| | | | 1101 |
| | | 111 | 1110 |
| | | | 1111 |

A complete code is one that has

$$\sum_i 2^{-l_i} = 1.$$

$$L(C,X) = \sum_i p_i l_i$$

Define ideal length

$$l_i^* = \log_2 \frac{1}{p_i} = h_i$$

$l_i$

$$C(X) \equiv \sum_i p_i l_i$$

$$l_i^* = \log_2 \frac{1}{p_i} = h_i$$

ideal length

$$l_i \longleftrightarrow q_i$$

IMPLICIT PROBS

$$q_i \equiv \frac{2^{-l_i}}{Z}$$

$n_i$

where $Z = \sum_i$

Note

$Z \leq 1$ for any UD code

$Z = 1$ for complete code

PROOFS

$-l_i$

$Z$

$$l_i = \log_2 \frac{1}{q_i} - \log_2 z$$

$$L(C,X) = \sum_i p_i \left[ \log_2 \frac{1}{q_i} - \log_2 z \right]$$

$$\log_2 z$$

$$- p_i \left[ \log_2 \frac{1}{q_i} - \log_2 z \right]$$

$$\sum_i p_i \log_2 \frac{1}{p_i} + \sum_i p_i \log \frac{p_i}{q_i} - \log z$$

Gibbs' inequality

$$\sum p \log \frac{p}{q} \geq 0$$

with equality only if $q = p$

$$\sum_i p_i \log \frac{p_i}{q_i} - \log z$$

$$+ D_{KL}(p \| q)$$

Gibbs inequality

$$\sum p \log \frac{p}{q} \geq 0$$

with equality only if $q = p$

$$-\log z$$

$\geq 0$ if code is UD,
with equality only if
the code is complete

$$L(C,X) \geq H(X)$$

equality iff

$$l_i = \log_2 \frac{1}{p_i}$$

& complete code

y if
e is complete

$$l_i^* = \log_2 \frac{1}{p_i}$$

Not integers

$$H(X) \leq L < H(X) + 1$$

# OPTIMAL SYMBOL CODES!

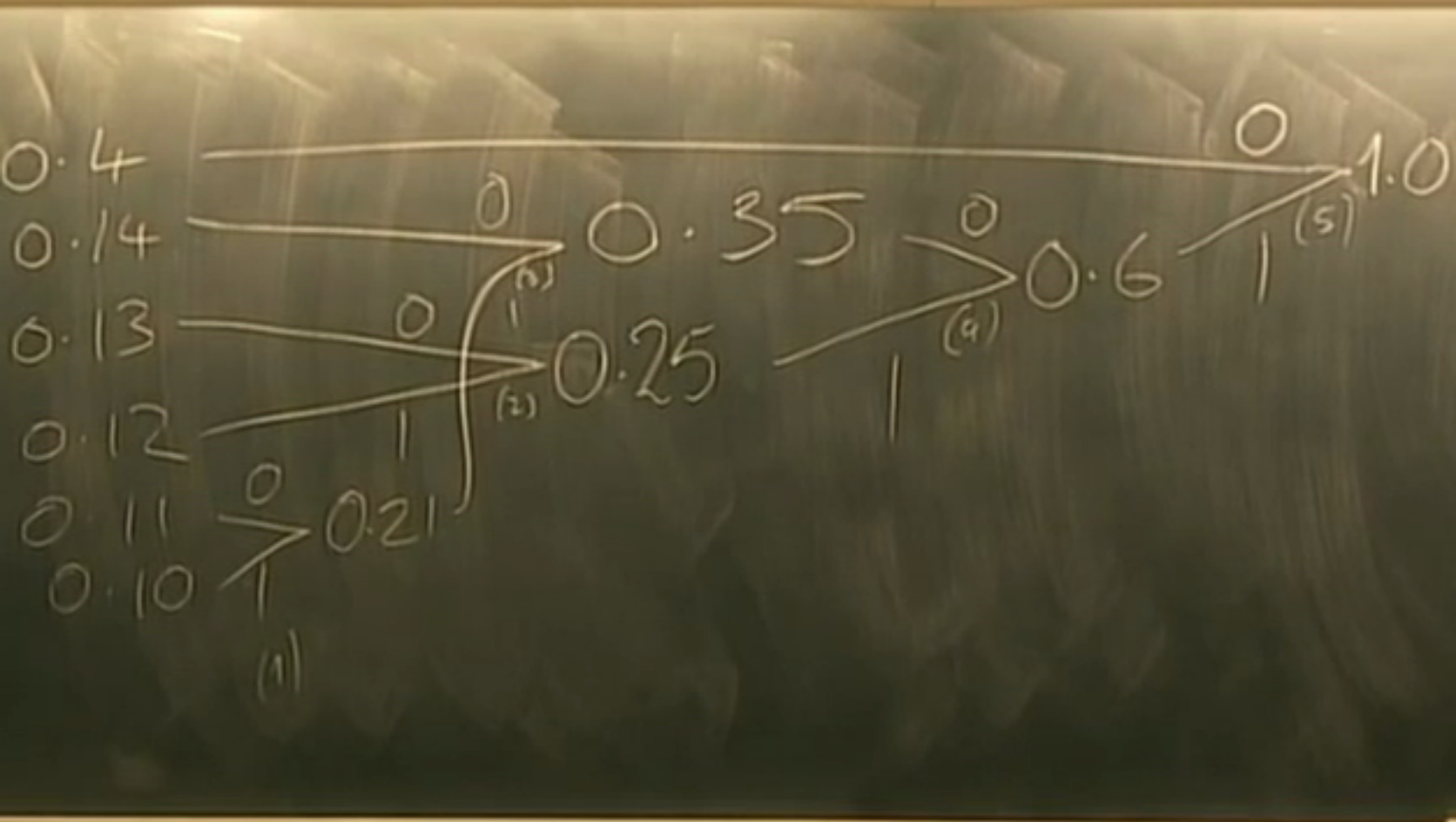
Huffman algorithm.

SUMMARY build binary tree, starting from the (furthest) leaves

ALGM

combine
the
2 symbols
w/ smallest
probability

& repeat


0.4
0.14
0.13
0.12
0.11
0.10
0
0
0
1
0
1
0.35
0.25
0.21
(3)
(2)
(1)
0
1
0.6
(4)
0
1
1.0
(5)

# Gibbs inequality

$$\sum_i p_i \log \frac{p_i}{q_i} \geq 0$$

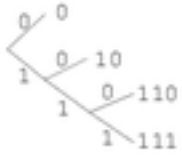
[equality only if $\mathbf{q} = \mathbf{p}$]

# Symbol code summary

- unique decodeability

$$\text{— requires } \sum_i 2^{-l_i} \leq 1 \quad \text{Kraft inequality}$$

[if equality holds, the code is 'complete']



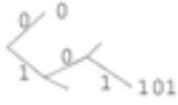

- prefix codes
  - and binary trees




- whenever a code achieved L=H, it had

codelengths $l_i^*$ equal to the information contents

$$l_i^* = \log \frac{1}{p_i}$$




- optimal symbol codes

The optimal symbol code's expected length $L$ satisfies

$$H(X) \leq L < H(X) + 1$$

- Huffman algorithm

# Results stated without proof (see Chapter 5)

- Kraft inequality

$$\sum_i 2^{-l_i} \leq 1$$

- Gibbs inequality

$$\sum_i p_i \log \frac{p_i}{q_i} \geq 0$$

- Right hand side of the symbol-code source coding theorem

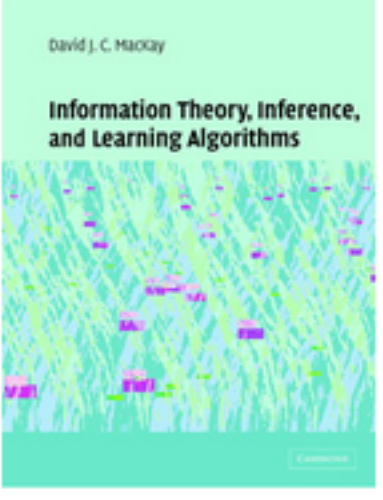
$$H(X) \leq \overbrace{L(C,X) < H(X)+1}$$

- Huffman coding is optimal

Project:

Invent a compressor and uncompressor for a source file of $N = 10,000$ bits, each having probability $f = 0.01$ of being a 1.
Implement them and/or
estimate how well your method works.

- Other recommended exercises: 5.22, 5.26, 5.27
- Reading: Chapters 1, 2, 4, 5.
- Advance reading: Chapters 8, 9, 10




Information Theory, Inference, and Learning Algorithms
Cambridge University Press (2003). Available free online